Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Cubic functions

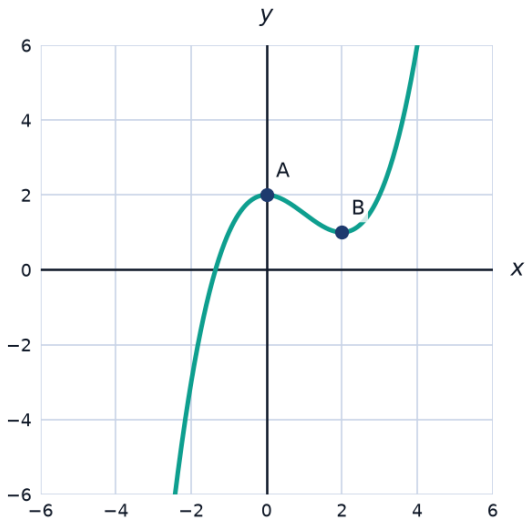
Consider a cubic polynomial $P(x) = ax^3 + bx^2 + cx + d$. Given that $P(1) = 10$, $P(2) = 27$, $P(3) = 58$, and $P(4) = 115$, determine the value of a.

2 [Foundational] • Target: Power functions

Prove by direct deductive reasoning that the difference between the squares of any two odd integers is always divisible by 8.

3 [Foundational] • Target: Direct variation

The sum of the first n terms of an arithmetic series is given by $S_n = 4n^2 - 3n$. (a) Find the first term u_1 and the second term u_2. (b) Find the common difference d. (c) Find an explicit expression for the nth term u_n, and calculate u_{25}.



4 [Foundational] • Target: Hyperbolas and inverse variation

A particle's position on the x-axis at time t (in seconds) is given by the function $x(t) = t^3 - 6t^2 + 9t + 1$. The particle starts at $t = 0$. Determine the time intervals during which the particle is moving in the negative direction and calculate the total distance traveled by the particle during the first 4 seconds.

5 [Foundational] • Target: Transformations of hyperbolas

Consider a geometric progression where the sum of the first n terms is given by $S_n = \frac{3(2^n - 1)}{2}$. A second sequence, denoted by a_n, is defined such that $a_n = S_n - S_{n-1}$ for $n \geq 2$, and $a_1 = S_1$. Find the value of $a_5 - a_3$.

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6 [Foundational] • Target: Graphing square root functions

In triangle ABC, side $a = 7$ cm, side $b = 9$ cm, and angle $A = 42^\circ$. (a) Show that two distinct triangles can be formed with these given measurements. (b) Calculate both possible values for angle B and the corresponding values of side c.

7 [Foundational] • Target: Cubic functions

Three mobile antennas are located at coordinates $A(0, 0)$, $B(6, 0)$, and $C(2, 4)$. (a) Find the equation of the perpendicular bisector of segment AB. (b) Find the equation of the perpendicular bisector of segment AC. (c) Find the coordinates of the Voronoi vertex (circumcentre) equidistant from all three antennas, and calculate the radius of the circumcircle.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Power functions

Six wines were evaluated by two independent sommeliers, X and Y, giving the following ratings: Wine: A, B, C, D, E, F Judge X: 88, 72, 95, 60, 81, 75 Judge Y: 84, 78, 91, 65, 85, 70 (a) Construct a table of ranks for both judges. (b) Calculate Spearman's rank correlation coefficient r_s. (c) State what this coefficient indicates about the judges' agreement.

9 [Intermediate] • Target: Direct variation

A small business borrows USD 40,000 to renovate its premises. The loan carries an annual nominal interest rate of 7.2% compounded monthly, with equal monthly repayments made at the end of each month over 6 years. (a) Calculate the monthly repayment amount. (b) Calculate the total amount repaid and the total interest charged on the loan. (c) After making payments for 3 full years, determine the outstanding loan balance.

10 [Intermediate] • Target: Hyperbolas and inverse variation

A particle moves along a straight line such that its velocity in m/s at time t seconds is given by $v(t) = 6t - 2t^2$ for $0 \le t \le 4$. (a) Find the acceleration of the particle at $t = 2$ s. (b) Determine the net displacement of the particle between $t = 0$ and $t = 4$. (c) Calculate the total distance travelled by the particle over the interval $[0, 4]$.

11 [Intermediate] • Target: Transformations of hyperbolas

A curve has equation $f(x) = \frac{2x^2}{x^2 + k}$ for some positive constant $k > 0$. (a) Write down the horizontal asymptote of $y = f(x)$. (b) Find $f'(x)$ in terms of k. (c) Given that the gradient of the tangent to the curve at $x = 1$ is equal to $\frac{1}{2}$, find all exact possible values of k.

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12 [Intermediate] • Target: Graphing square root functions

A closed cylindrical metal container with circular base of radius r cm and height h cm has a fixed volume of 250π cm^3. (a) Express h in terms of r. (b) Show that the total surface area is $A(r) = 2\pi r^2 + \frac{500\pi}{r}$. (c) Find the radius r that minimizes the surface area, and show that for this optimal cylinder, the height equals the diameter ($h = 2r$).

13 [Intermediate] • Target: Cubic functions

A continuous random variable X is normally distributed with mean μ and standard deviation σ. Given that $P(X < 45) = 0.12$ and $P(X > 85) = 0.08$, find the value of μ and σ correct to 3 significant figures.

14 [Intermediate] • Target: Power functions

Consider the function $f(x) = x^2 e^{-x}$. (a) Find $f'(x)$ and express it in factored form. (b) Hence, find the coordinates of all stationary points of f. (c) Use the second derivative test to classify each stationary point as a local maximum or local minimum.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Direct variation • Cross-Lesson: 1.01 Number & Algebra

Find the exact area of the finite region enclosed between the parabola $y = 3x - x^2$ and the line $y = x - 3$.

16 [Challenging] • Target: Hyperbolas and inverse variation • Cross-Lesson: 3.01 Geometry & Trigonometry

In triangle PQR, angle PQR = 90 degrees. S is a point on PR such that QS is perpendicular to PR. The length of PQ is 15 cm and the length of QR is 20 cm. Let the length of PS be x cm and the length of SR be y cm. Find the value of x.

17 [Challenging] • Target: Transformations of hyperbolas • Cross-Lesson: 4.01 Statistics & Probability

The lifespan X in hours of an electronic component is normally distributed with mean $\mu = 1200$ hours and standard deviation $\sigma = 150$ hours. (a) Find the probability that a randomly selected component lasts more than 1350 hours. (b) In a batch of 500 components, estimate how many will last between 1000 and 1400 hours. (c) The manufacturer guarantees the top 10% longest-lasting components. Determine the minimum lifespan required to receive this guarantee.

18 [Challenging] • Target: Graphing square root functions • Cross-Lesson: 5.01 Calculus

In a factory, 60% of components are manufactured by Machine A and 40% by Machine B. It is known that 3% of components produced by Machine A are defective, whereas 5% of components produced by Machine B are defective. (a) Find the probability that a randomly chosen component is defective. (b) Given that a randomly chosen component is defective, find the probability that it was produced by Machine B. (c) Determine whether the events "component produced by Machine B" and "component is defective" are independent.

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19 [Challenging] • Target: Cubic functions • Cross-Lesson: 6.01 AI SL — Specific Topics

In a 2D Cartesian plane, points A, B, C have coordinates $A(2, -1)$, $B(5, 3)$, and $C(-1, k)$. (a) Find the vectors $\vec{AB}$ and $\vec{AC}$ in component form. (b) Find the value of k for which $\vec{AB}$ is perpendicular to $\vec{AC}$. (c) For $k = 3$, calculate the exact cosine of the angle between $\vec{AB}$ and $\vec{AC}$.

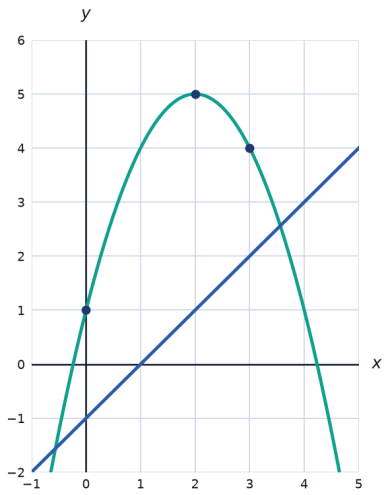
20 [Challenging] • Target: Power functions • Cross-Lesson: 1.01 Number & Algebra

Consider the binomial expansion of $\left(x^2 - \frac{2}{x}\right)^9$. (a) Write down the general term in terms of r. (b) Find the coefficient of x^3. (c) Find the term independent of x.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Cubic functions

For the model $f(x) = 2x + 6$ in cubic functions, evaluate $f(2)$.



a. 10

b. 12

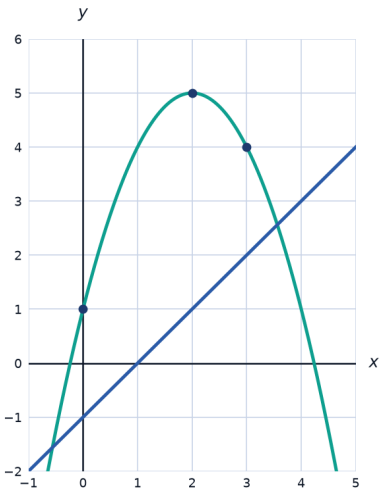
c. 4

d. 14

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22 [Foundational] • Target: Cubic functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

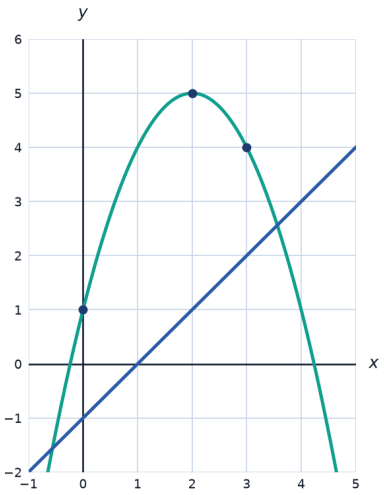
b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

23 [Foundational] • Target: Cubic functions

For the model $f(x) = 2x + 6$ in cubic functions, evaluate $f(5)$.



a. 16

b. 18

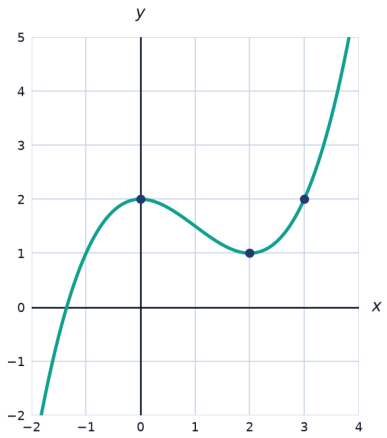
c. 10

d. 20

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24 [Foundational] • Target: Power functions

For the model $f(x) = 3x + 7$ in power functions, evaluate $f(2)$.



a. 13

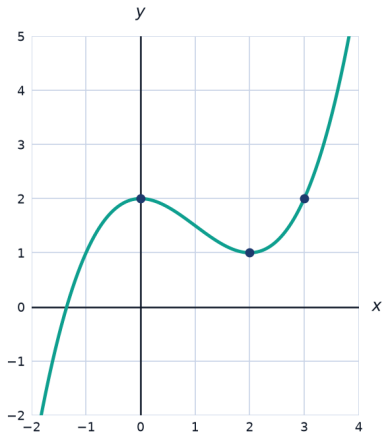
b. 16

c. 6

d. 17

25 [Foundational] • Target: Power functions

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

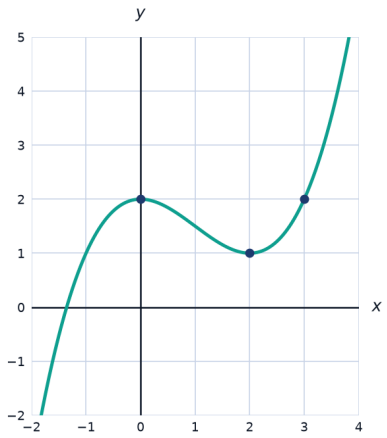
c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

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26 [Foundational] • Target: Power functions

For the model $f(x) = 3x + 7$ in power functions, evaluate $f(5)$.



a. 22

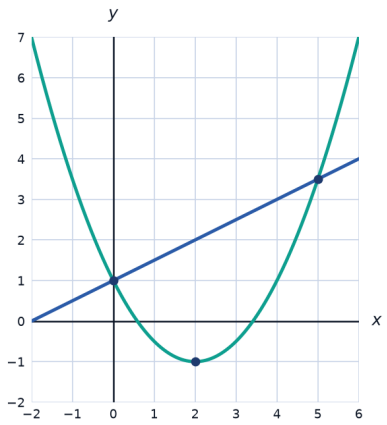
b. 25

c. 15

d. 26

27 [Foundational] • Target: Direct variation

For the model $f(x) = 4x + 8$ in direct variation, evaluate $f(2)$.



a. 16

b. 20

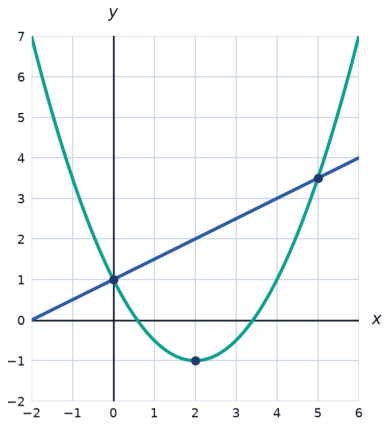
c. 8

d. 20

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28 [Foundational] • Target: Direct variation

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cubic functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

30 [Intermediate] • Target: Cubic functions

A data set recorded for cubic functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

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31 [Intermediate] • Target: Cubic functions

For the model $f(x) = 2x + 6$ in cubic functions, evaluate $f(4)$.

a. 14

b. 16

c. 8

d. 18

32 [Intermediate] • Target: Cubic functions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

33 [Intermediate] • Target: Power functions

A data set recorded for power functions consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

34 [Intermediate] • Target: Power functions

For the model $f(x) = 3x + 7$ in power functions, evaluate $f(3)$.

a. 16

b. 19

c. 9

d. 20

35 [Intermediate] • Target: Power functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

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36 [Intermediate] • Target: Direct variation

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

37 [Intermediate] • Target: Direct variation

For the model $f(x) = 4x + 8$ in direct variation, evaluate $f(3)$.

a. 20

b. 24

c. 12

d. 24

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Direct variation • Cross-Lesson: 5.01 Calculus

For the model $f(x) = 4x + 8$ in direct variation, evaluate $f(4)$.

a. 24

b. 28

c. 16

d. 28

39 [Challenging] • Target: Direct variation • Cross-Lesson: 6.01 AI SL — Specific Topics

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

40 [Challenging] • Target: Hyperbolas and inverse variation • Cross-Lesson: 1.01 Number & Algebra

A data set recorded for hyperbolas and inverse variation consists of the four values $\{14, 18, 22, 28\}$.

What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

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41 [Challenging] • Target: Hyperbolas and inverse variation • Cross-Lesson: 3.01 Geometry & Trigonometry

For the model $f(x) = 5x + 9$ in hyperbolas and inverse variation, evaluate $f(4)$.

a. 29

b. 34

c. 20

d. 33

42 [Challenging] • Target: Hyperbolas and inverse variation • Cross-Lesson: 4.01 Statistics & Probability

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

43 [Challenging] • Target: Transformations of hyperbolas • Cross-Lesson: 5.01 Calculus

A data set recorded for transformations of hyperbolas consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

44 [Challenging] • Target: Transformations of hyperbolas • Cross-Lesson: 6.01 AI SL — Specific Topics

For the model $f(x) = 6x + 3$ in transformations of hyperbolas, evaluate $f(4)$.

a. 27

b. 33

c. 24

d. 31

45 [Challenging] • Target: Graphing square root functions • Cross-Lesson: 1.01 Number & Algebra

For the model $f(x) = 2x + 4$ in graphing square root functions, evaluate $f(3)$.

a. 10

b. 12

c. 6

d. 14

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